

STAT 6410

Homework 5

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1. Problem 2.15(c)

Consider the probability space $((0, 1), \mathcal{B}((0, 1)), m)$, where $m(\cdot)$ is the Lebesgue measure.

(c) Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a cdf. For $0 < x < 1$, let

$$F_1^{-1}(x) = \inf\{y : y \in \mathbb{R}, F(y) \geq x\}$$

$$F_2^{-1}(x) = \sup\{y : y \in \mathbb{R}, F(y) \leq x\}$$

Let Z_i be the random variable defined by $Z_i = F_i^{-1}(x)$, $0 < x < 1$, $i = 1, 2$.

(i) Find the cdf of Z_i , $i = 1, 2$

(Hint: Verify using the right continuity of F that for any $0 < x < 1$, $t \in \mathbb{R}$,
 $F(t) \geq x \iff F_1^{-1}(x) \leq t$.)

(ii) Show also that $F_1^{-1}(\cdot)$ is left continuous and $F_2^{-1}(\cdot)$ is right continuous.

Ans: (i)

Finding the cdf of Z_1

Proof. Let $t \in \mathbb{R}$ and $x \in (0, 1)$ be given.

Part 1: WTS $F(t) \geq x \implies F_1^{-1}(x) \leq t$.

Suppose $F(t) \geq x$.

$$F_1^{-1}(x) = \inf\{y : y \in \mathbb{R}, F(y) \geq x\}.$$

From the definition of $F_1^{-1}(\cdot)$, we know that there exists no $r \in \mathbb{R}$ such that $F(r) \geq x$ and $r \leq F_1^{-1}(x)$.

Thus, since $x \leq F(t)$, we know that, $F_1^{-1}(x) \leq t$.

Part 2: WTS $F(t) \geq x \iff F_1^{-1}(x) \leq t$.

Suppose $F_1^{-1}(x) \leq t$.

Define $\{y_n\}_{n \geq 1}$ to be a decreasing sequence from $\{y : y \in \mathbb{R}, F(y) \geq x\}$.

By the right continuity of $F(\cdot)$ (F is a cdf), $\lim_{n \rightarrow \infty} y_n = F_1^{-1}(x)$.

$$t \geq F_1^{-1}(x) \implies F(t) \geq F(F_1^{-1}(x)) = F(\lim_{n \rightarrow \infty} y_n) = \lim_{n \rightarrow \infty} F(y_n) \geq x$$

$$\therefore F(t) \geq x \iff F_1^{-1}(x) \leq t.$$

$$\therefore F(t) \geq x \iff F_1^{-1}(x) \leq t.$$

$$\begin{aligned} P(Z_1 \leq t) &= P(F_1^{-1}(x) \leq t) \\ &= P(x \leq F(t)) \quad (\text{since } F(t) \geq x \iff F_1^{-1}(x) \leq t) \\ &= m(\{x : x \leq F(t)\}) = F(t) \end{aligned}$$

Thus, the cdf of Z_1 is $F(t)$ for all $t \in \mathbb{R}$.

Finding the cdf of Z_2

Using a similar logic, we can show that for all $x \in (0, 1)$ $F(t) > x \implies F_2^{-1}(x) \leq t$. This means that $F(t) = m(\{x : F(t) > x\}) \leq m(\{F_2^{-1}(x) \leq t\})$. We can, in a very similar manner to the first part of this problem show that $F(t) > x \iff F_2^{-1}(x) \leq t$. This means that $m(\{F_2^{-1}(x) \leq t\}) \leq m(\{x : F(t) > x\}) = F(t)$. Thus, $F(t) = m(\{F_2^{-1}(x) \leq t\})$

$$P(Z_2 \leq t) = m(\{F_2^{-1}(x) \leq t\}) = F(t)$$

Thus, the cdf of Z_2 is $F(t)$ for all $t \in \mathbb{R}$.

(ii)

also use the hint in (ii)

Proof. Part 1: WTS $F_1^{-1}(\cdot)$ is left continuous

Let $\epsilon > 0$ be given. There exists δ_ϵ such that $F(F_1^{-1}(x) - \epsilon) < x - \delta_\epsilon$. From $F(t) \geq x \iff F_1^{-1}(x) \leq t$ (in part (i)), we know that $F_1^{-1}(x - \delta_\epsilon) > F_1^{-1}(x - \epsilon)$.

Define $x^* = (x - \delta_\epsilon, x)$.

$$\begin{aligned} F_1^{-1}(x - \delta_\epsilon) &< F_1^{-1}(x^*) \implies \\ F_1^{-1}(x) - F_1^{-1}(x^*) &< F_1^{-1}(x) - F_1^{-1}(x - \delta_\epsilon) < \epsilon \end{aligned}$$

Thus, $F_1^{-1}(\cdot)$ is left continuous.

Part 2: WTS $F_2^{-1}(\cdot)$ is right continuous

Again, referring back to (i): for all $\epsilon > 0$, there exists some $\delta_\epsilon > 0$ such that $F(F_2^{-1}(x + \epsilon)) > x + \delta_\epsilon \implies F_2^{-1}(x + \delta_\epsilon) \leq F_2^{-1}(x + \epsilon)$. Thus, $F_2(\cdot)$ is right continuous. $\square$

$\square$

2. Problem 2.20

Apply Corollary 2.3.5 to show that for any collection $\{a_{ij} : i, j \in \mathbb{N}\}$ of nonnegative

numbers,

$$\sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) = \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right).$$

Corollary 2.3.5: *Let $\{h_n\}_{n \geq 1}$ be a sequence of nonnegative measurable functions on a measure space $(\Omega, \mathcal{F}, \mu)$. Then*

$$\int \left(\sum_{n=1}^{\infty} h_n \right) d\mu = \sum_{n=1}^{\infty} \int h_n d\mu$$

Ans:

Proof. Define $\mu(\cdot)$ to be a counting measure.

If $\{b_k : k \in \mathbb{N}\}$ is a collection of nonnegative numbers:

$$\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} b_k I(\{k\}) = \int b_k d\mu(\{i\})$$

We can apply this to our problem at hand:

$$\begin{aligned} \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) &= \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) I(\{i\}) = \int \left(\sum_{j=1}^{\infty} a_{ij} \right) d\mu(\{i\}) \\ &= \sum_{j=1}^{\infty} \left(\int a_{ij} d\mu(\{i\}) \right) && \text{(by Corollary 2.3.5)} \\ \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} I(\{i\}) \right) &= \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \times 1 \right) \\ &= \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right) \end{aligned}$$

□

3. Problem 2.25

If $f(x) = I_{\mathbb{Q}_1}(x)$ where $\mathbb{Q}_1 = \mathbb{Q} \cap [0, 1]$, $\mathbb{Q}$ being the set of all rational, then show that for any partition P , $U(P, f) = 1$ and $L(P, f) = 0$.

Ans:

Proof. Define a partition, P , on $\mathbb{Q}_1$ as $P = (x_0 = 0, x_1, \dots, x_n = 1)$

Rationals are dense in the real numbers, thus for all $0 \leq i \leq 1$, there exists a rational number in $[x_i, x_{i+1}]$.

Thus, $\sup(f(x) : x_i \leq x \leq x_{i+1}) = 1$ for all i .

$$U(P, f) = \sum_{i=1}^n \sup(f(x) : x_i \leq x \leq x_{i+1})(x_{i+1} - x_i) = \sum_{i=1}^n 1(x_{i+1} - x_i) = \sum_{i=1}^n (x_{i+1} - x_i) = 1 - 0 = 1.$$

Similarly, irrational numbers are dense in the real numbers, thus for all $0 \leq i \leq 1$, there exists an irrational number in $[x_i, x_{i+1}]$.

Thus, $\inf(f(x) : x_i \leq x \leq x_{i+1}) = 0$ for all i .

$$L(P, f) = \sum_{i=1}^n \inf(f(x) : x_i \leq x \leq x_{i+1})(x_{i+1} - x_i) = \sum_{i=1}^n 0(x_{i+1} - x_i) = 0. \quad \square$$

4. Extra Problem (A)

Let $f : \Omega \rightarrow \mathbb{R}$ be a $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable *nonnegative* function. Define for $n \geq 1$, and $x \in \Omega$

$$f_n(x) = \begin{cases} \frac{(i-1)}{2^n}, & \text{if } \frac{(i-1)}{2^n} \leq f(x) < \frac{i}{2^n} \text{ for } i = 1, \dots, n2^n \\ n, & \text{if } f(x) \geq n \end{cases}$$

Then prove for the following:

- (i) For each $n \geq 1$, $f_n(x)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
- (ii) For each $x \in \Omega$, $f_n(x)$ is increasing with n (i.e. $\{f_n(\cdot)\}$ is an increasing sequence of functions).
- (iii) Prove that $f_n(x) \rightarrow f(x)$ pointwise (i.e. for each fixed x), as $n \rightarrow \infty$

Ans: (i)

Proof. Let $n \geq 1$ be given.

We can write $f_n(x) = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} I(\frac{(i-1)}{2^n} \leq f(x) < \frac{i}{2^n}) + nI(f(x) \geq n)$

$$= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} I(x : x \in f^{-1}([\frac{(i-1)}{2^n}, \frac{i}{2^n}))) + nI(x : x \in f^{-1}([n, \infty)))$$

$$[0, \frac{1}{2^n}), [\frac{1}{2^n}, \frac{2}{2^n}), \dots, [\frac{n2^n-1}{2^n}, n), [n, \infty) \in \mathcal{B}(\mathbb{R})$$

Thus, for all $i \in \{1, 2, \dots, n2^n\}$, $\{x : x \in f^{-1}([\frac{(i-1)}{2^n}, \frac{i}{2^n}))\} \in \mathcal{F}$ and

$\{x : x \in f^{-1}([n, \infty))\} \in \mathcal{F}$

Thus, $f_n(x)$ is a sum of indicator functions multiplied by constants.

All of the indicator functions are $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable, since all of the sets by which the indicator function is defined are elements of $\mathcal{F}$.

By **Corollary 2.1.4**, $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable functions are closed under addition and scalar multiplication, $f_n(x)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. $\square$

(ii)

Proof. Let $x \in \Omega$ be given.

First, let's rewrite $f_n(x)$ to make it easier to work with:

$$\begin{aligned} f_n(x) &= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} I\left(\frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}\right) + nI(f(x) \geq n) \\ &= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} I((i-1) \leq 2^n f(x) < i) + nI(f(x) \geq n) \\ &= \frac{\lfloor \min(2^n f(x), 2^n n) \rfloor}{2^n} \end{aligned}$$

Now to answering the question at hand:

$$\begin{aligned} f_{n+1}(x) &= \frac{\lfloor \min(2^{n+1} f(x), 2^{n+1}(n+1)) \rfloor}{2^{n+1}} \\ &= \frac{\lfloor 2 \min(2^n f(x), 2^n(n+1)) \rfloor}{2^{n+1}} = \frac{2 \lfloor \min(2^n f(x), 2^n(n+1)) \rfloor}{2^{n+1}} \\ &= \frac{\lfloor \min(2^n f(x), 2^n(n+1)) \rfloor}{2^n} \geq \frac{\lfloor \min(2^n f(x), 2^n n) \rfloor}{2^n} \\ &= f_n(x) \end{aligned}$$

□

(iii)

Proof. Let $x \in \Omega$ be given.

Thus, there exists some $N \in \mathbb{N}$ such that $N \geq f(x)$. Since we are investigating the behavior of n it approaches ∞ , $\min(2^n f(x), 2^n n) = 2^n f(x)$ as $n \rightarrow \infty$.

$$\begin{aligned} \lim_{n \rightarrow \infty} |f_n(x) - f(x)| &= \lim_{n \rightarrow \infty} \left| \frac{2^n f(x) - \lfloor \min(2^n f(x), 2^n n) \rfloor}{2^n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{|2^n f(x) - \lfloor 2^n f(x) \rfloor|}{2^n} \leq \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0 \end{aligned}$$

Thus, $f_n(x) \rightarrow f(x)$ point-wise as $n \rightarrow \infty$ for all $x \in \Omega$

□

5. Extra Problem (B)

If $f, g \in L_1(\Omega, \mathcal{F}, \mu)$,

show

$$\min(f, g) \in L_1(\Omega, \mathcal{F}, \mu)$$

and

$$\min \left(\int f d\mu, \int g d\mu \right) \geq \int \min(f, g) d\mu.$$

Ans: WTS(Want to Show): $\min(f, g) \in L_1(\Omega, \mathcal{F}, \mu)$

Proof. $|\min(f, g)| \leq |f| + |g|$

Since, $f, g \in L_1(\Omega, \mathcal{F}, \mu)$, $\int |f| d\mu < \infty$ and $\int |g| d\mu < \infty$.

This means that $\int |f| + |g| d\mu < \infty$. By monotonicity, $\int |\min(f, g)| d\mu \leq \int |f| + |g| d\mu < \infty$.

Thus, $\min(f, g) \in L_1(\Omega, \mathcal{F}, \mu)$ □

WTS: $\min \left(\int f d\mu, \int g d\mu \right) \geq \int \min(f, g) d\mu$

Proof. First, we will state this equality: $\min(x, y) = \frac{1}{2}(x + y - |x - y|)$.

$$\begin{aligned} \int \min(f, g) d\mu &= \frac{1}{2} \int f + g - |f - g| d\mu \\ &= \frac{1}{2} \left(\int f d\mu + \int g d\mu - \int |f - g| d\mu \right) \\ &\leq \frac{1}{2} \left(\int f d\mu + \int g d\mu - \left| \int f - g d\mu \right| \right) \quad (*) \\ &= \frac{1}{2} \left(\int f d\mu + \int g d\mu - \left| \int f d\mu - \int g d\mu \right| \right) \\ &= \min \left(\int f d\mu, \int g d\mu \right) \end{aligned}$$

(*) Since f and g are measurable, $f - g$ is clearly measurable. Thus $\int |f - g| d\mu \geq \left| \int f - g d\mu \right|$ □